Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/03/2016
PI Name:	Irudayaraj, Joseph Ph.D.		
Project Title:	Defining Epigenetic Programming During Flight Expeditions In Differentiating Embryonic Stem Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	josephi@purdue.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-0388
Organization Name:	Purdue University		
PI Address 1:	Agricultural and Biological Engineering		
PI Address 2:	225 S University St, ABE 215 Building		
PI Web Page:			
City:	West Lafayette	**State:**	IN
Zip Code:	47907-2093	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
Contact Email:	elizabeth.taylor-23@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2015 per A. Chu; original end date was 7/31/2013 (Ed., 7/14/14) NOTE: Extended to 7/31/14 per A. Chu/ARC (Ed., 7/22/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lossie, Amy (Purdue University)		
Grant/Contract No.:	NNX12AN09G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Gene expression during cellular differentiation is precisely choreographed. Its timing and location controls cell fate; early occurring mistakes are readily amplified during cell division, resulting in birth defects, disease, or death. Epigenetic marks constitute chemical modifications that are added 'on top of' DNA that direct gene expression (turn genes ON/OFF) without causing a genetic mutation. These marks include DNA methylation (the 5-methylcytosine, 5mC), and hydroxymethylation (the 5-hydroxymethylcytosine, 5hmC) and histone tail modifications (methylation, acetylation, etc). These heritable, yet reversible instructions are affected by many environmental factors, including diet, stress, alcohol, radiation, environmental contaminants, and cell culture conditions. In space "the final frontier", organisms are exposed to powerful environmental forces (microgravity, cosmic radiation, magnetic fields) that potentially govern epigenetic gene regulation. These alter cell fate by modulating the epigenome leading to changes in gene expression. While a majority of the existing work describes the 5mC and histone modification landscapes in Embryonic Stem Cells (ESCs), limited data exist on epigenetic programming in embryoid bodies and only a handful of studies have documented the 5hmeC landscape in pluripotent ESCs. Furthermore, the effects of space flight on 5meC, 5hmeC, and histone modifications remain unexplored. However, preliminary studies suggest that space flight inhibits differentiation of mouse ESCs, arguing that these environmental forces negatively impact epigenetic programming, leading to sustained ESC pluripotency. In this study, we investigate the effects of simulated microgravity on 5mC and 5hmC patterns in the genome. To our knowledge, this study is the first to document the effects of simulated microgravity on epigenetic regulation of gene expression. The following methods were developed to complete the Next Generation Sequencing (NGS) experiments for the two cell lines. By elucidating the mechanisms that govern epigenetic programming and gene regulation in single cells, we will provide fundamental knowledge that will broadly impact the epigenetics community, basic biology, and NASA's space biology programs. The goal of our interdisciplinary effort is to develop an integrated approach comprised of Next Generation Sequencing and single cell microscopy platforms to understand how the space environment affects DNA methylation, hydroxymethylation, and histone modifications to regulate gene expression during differentiation using an Embryonic Stem Cell model. The two key aims of the grant were: Aim 1. Determine the effects of simulated microgravity on epigenetic modifications and gene expression in lymphoblastoid cells. Aim 2. Correlate DNA methylation (5mC and 5hmC) and histone modifications in single cells at specified loci with gene expression and ESC differentiation following exposure to the space flight environment. Based on the reviewers' comments Aim 1 was modified to demonstrate proof-of-concept for the initial Phase funding to test the hypothesis that microgravity in fact alters DNA methylation. Revised Aim 1. Determine the effects of space flight on epigenetic modifications and gene expression in HL-60 and TK6 cells. We hypothesize that microgravity and other environmental conditions during space flight alter DNA methylation, leading to changes in the expression of specific genes that are impacted by this unique environment. We will assess 5hmeC and 5meC using next generation sequencing and correlate these epigenetic marks with the expression of five key genes sensitive to ground conditions (static and rotary cultures) and space flight environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will constitute one of the first efforts to assess how environmental conditions (i.e., microgravity, radiation) affect epigenetic patterns (i.e., DNA methylation and hydroxymethylation patterns and histone modifications) to coordinate gene expression. We expect these epigenetic events to explain genetic aberrations that cannot be explained by conventional genetic studies. Discovery of site-specific alteration of the epigenome and its role in gene expression will be highly significant not only to the field of epigenetics, but to biology in general. By elucidating epigenetic consequences due to the space flight environment in pluripotent embryonic stem cells (ESCs) and their differentiated counterparts, embryoid bodies, we will begin to discover mechanistically how gravity alters epigenetic modifications and in turn, gene expression in the three basic germlayers (endoderm, ectoderm, and mesoderm) during cellular differentiation. Our ability to complement genomic sequencing with single cell tools will facilitate exploration of any gene at single cell resolution from any organism, a heretofore, elusive feature. With slight modifications, we will have the ability to assess DNA binding factors, reveal functional irregularities based on epigenetic profiles, and screen for drugs that target specific enzymes that add or remove epigenetic modifications. Our methodologies can also be used to follow chromatin through the cell cycle and to determine how DNA damage triggers changes in chromatin. Key applications also include defining real-time chromatin dynamics during cellular growth, differentiation, and early development not only in response to space flight but also diet, nutrition, and alcohol, among other environmental factors. We expect our contributions to impact space biology research in NASA, and the planning of future long-term space flight expeditions.
Task Progress:	The task has been completed. HL60 and TK cells were grown in the respective conditions and subjected to Next Generation Sequencing (NGS) analysis to determine the methylated regions so that we can enhance our understanding of the gene expression under microgravity conditions. Deliverables of the study that have been accomplished include: 1) Development and optimization of methods for Next Generation Sequencing (MeDIP; hMeDIP) of the two cell lines (ie. HL60 and TK6) 2) Determination of the methylation and hydroxymethylation profiles in the genome and 3) Determination of the key genes that are undergoing change in gene transcription and are significantly methylated/hydroxymethylated by comparative bioinformatics approach on RNA-seq and MeDIP-seq/hMeDIP-seq data. List of Publications 1. Chowdhury, B., Arun, G., Thimmapuram, J., and Irudayaraj, J. 2015. A study of alterations in DNA epigenetic modifications (5mC and 5hmC) and gene expression influenced by simulated microgravity in human lymphoblastoid cells. Plos One, 11(1), e0147514. 2. Chowdhury, B., Chen, J., Liu, S., Lossie, A., and Irudayaraj, J. 2015. The hypomethylating agent Decitabine causes a paradoxical increase in 5-hydroxymethylcytosine in human leukemia cell. Nature Scientific Reports. 9281 (2015); doi :10.1038/srep09281 3. Chowdhury, B., Kirchmaier, A., Thimmapuram, J., Arun, G., and Irudayaraj, J. Kirchmaier, A. Sequencing of

	5-Hydroxymethylcytosine during myeloid differentiation in HL60 cells. BMC Genetics. (In Review)
Bibliography Type:	Description: (Last Updated: 06/03/2016)
Articles in Peer-reviewed Journals	Chowdhury B, Seetharam A, Wang Z, Liu Y, Lossie AC, Thimmapuram J, Irudayaraj J. "A study of alterations in DNA epigenetic modifications (5mC and 5hmC) and gene expression influenced by simulated microgravity in human lymphoblastoid cells." PLoS One. 2016 Jan 28;11(1):e0147514. http://dx.doi.org/10.1371/journal.pone.0147514 ; eCollection 2016. PubMed PMID: 26820575; PubMed Central PMCID: PMC4731572 , Jan-2016
Articles in Peer-reviewed Journals	Chowdhury B, McGovern A, Cui Y, Choudhury S, Cho I-L, Cooper B, Chevassut T, Lossie A, Irudayaraj J. "The hypomethylating agent Decitabine causes a paradoxical increase in 5-hydroxymethylcytosine in human leukemia cells." Scientific Reports. 2015 Apr 22;5:9281. http://dx.doi.org/10.1038/srep09281 ; PubMed PMID: 25901663. , Apr-2015
Articles in Peer-reviewed Journals	Chowdhury B, Kirchmaier A, Thimmapuram J, Arun G, Irudayaraj J, Kirchmaier A. "Sequencing of 5-Hydroxymethylcytosine during myeloid differentiation in HL60 cells." BMC Genetics. (In Review) As of March 2016. , Mar-2016